

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-I: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
< form action=" / register" method="POST">  
  < input type="text" name="name">  
  < input type=" email " name=" email">  
  <button>Submit</button>  
< / form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly. Questions:

- 1 . Identify appropriate HTML5 semantic elements missing in the structure.
- 2 Modify the structure to make it standards-compliant.
- 3 Interpret the HTTP request generated during form submission.
- 4 Analyze the HTTP response cycle from server to browser.

5. Suggest improvements for usability and accessibility.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet is used:

```
< html >
  <head>
    < title>Company</titIe>
  </head>
  <body>
    <div>
      <h1>WeIcome
      <p>About us
    < / div>
  </body>
< / html
>
```

Questions:

1. Identify improper nesting and missing closing tags.
2. Analyze how different browsers may interpret this structure.
3. Identify missing semantic elements (e.g., header, section).
4. Provide a corrected W3C-compliant HTML structure.
5. Suggest techniques to ensure cross-browser compatibility.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
< form action=" / submit" method="GET">
  < input type="text" name="username">
  < input type="password" name="password">
  <button type="button">Login</button>
< / form>
```

Questions:

1. Identify issues in the form configuration.
2. Analyze why the HTTP request is not triggered.
3. Interpret the role of GET method in this scenario.
4. Propose a corrected implementation.
5. Suggest improvements for secure data transmission.

Code of Conduct:

I K Manjunadha reddy/192372081 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K manju

RUBRICS FOR CALCULATION:

	htage	Level 4 – Exempla	Level 3 – Proficient	Level 2 – Developing	Beginnin
Understandi ng of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understandi ng with minor gaps	Basic understandi ng with some misconcepti on s	Poor understan din g; major misconce ptio ns
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well- structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organizatio n, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Answers

1)Online Symposium Portal

1. Identify missing HTML5 semantic elements

- <html>
- <head> o <title> •< main >
- <section>
- < label> for form fields

2. Standards-compliant HTML

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Symposium Registration</title>
</head>
<body>

<section>
<h2>Registration Form</h2>

<form action="/register" method="POST">
  <label for="name">Name</label>
  <input type="text" id="name" name="name" required>

```

```
<label for="email">Email</label>
<input type="email" id="email" name="email" required>

<button type="submit">Submit</button>
</form>

</section>
</main>
</body>
```

3. HTTP Request

- Method: POST ●
- URL: / register ● Data sent:

name=John

email=iohn@example.com

4. HTTP Response cycle

1. Browser sends POST request.
2. Server processes the data.
3. Server returns a response (e.g., 200 OK or 302 Redirect).
4. Browser displays confirmation page.

5. Usability & Accessibility Improvements

- Use < label> elements.
- Add required validation.
- Use semantic tags (<main>, <section>).
- Provide clear button text.
- Ensure keyboard accessibility.

2. Cross-Browser Compatibility Issue

1. Improper Nesting & Missing Closing Tags

- `<h1>` not closed.
- `<n>` not closed.
- Missing `<!DOCTYPE html>`.

2. Browser Interpretation

- Browsers may automatically close missing tags.
 - o Different browsers may display different layouts.

3. Missing Semantic Elements

- `<header>`
- `<main>`
- `<section>`

4. Correct W3C-Compliant HTML

```
    CTYPEhtml>  
<html lang="en">
```

```
<meta charset="UTF-8">  
<title>Company</title>  
</head>
```

```
<header>  
<h1>Welcome</h1>  
</header>
```

```
<main>  
<section>  
<p>About us</p>  
</section>  
</main>
```

```
</body>  
</html>
```

5. Cross-Browser Compatibility Techniques

- Use valid HTML5.
- Test on multiple browsers.
- Use CSS Reset/Normalize.
- Validate HTML and CSS. ● Use responsive design.

3. Form Submission Failure

1. Issues in Form Configuration

- Uses GET instead of POST.
- Button type is button " instead of submit".

2. Why HTTP Request is Not Triggered

- type= " button does not submit the form. ● Only type=" submit 'sends the request.

3. Role of GET Method

- Sends data in the URL.
- Suitable for search forms.
- Not secure for passwords.

4. Correct Implementation

```
<form action="/submit" method="POST">
```

```
<label>Username</label>
```

```
<input type="text" name="username" required>
```

```
<label>Password</label>
```

```
<input type="password" name="password" required>
```

```
<button type="submit">Login</button>
```

```
</form>
```

5. Secure Data Transmission Improvements

- Use POST method.
- Use HTTPS.
- Validate input on client and server.
- Hash passwords before storing.
- Enable authentication and session management.